

Semester: CE-5

Comsats University Islamabad Attock Campus



Submitted By: Sumaira Safeer
Registration number: FA22-BCE-019
Submitted To: Dr. Shujaat Ali Khan Tanoli
Course Name: Principle of Communication System
Program: BS Computer Engineering
Topic: Project Report

Department of Electrical and Computer Engineering

Reliable Digital Communication System for a Noisy Channel.

Table of Contents

1: Introduction.....	3
2: Literature Review.....	3
3: System Model	3
3.1.1: Binary Phase Shift Keying (BPSK)	7
3.1.2: Quadrature Phase Shift Keying (QPSK).....	7
3.1.3: 16-Quadrature Amplitude Keying (16-QAM).....	8
3.4: BER Analysis	9
3.4.1: BPSK Performance Over QPSK And 16-QAM	9
Derivation for BPSK:.....	9
3.4.2: QPSK with Coding:	12
3.4.3: 16-QAM with Coding:	12
4: Result And Discussion	12
4.1: Symbol Rate and Effective Data Rate	13
4.2: Calculating Symbol Rate and Effective Data Rate	13
4.2.1: BPSK	13
4.2.2: QPSK.....	14
4.2.3: 16-QAM.....	14
5: Simulation.....	14
6: Result and Discussion:.....	21
6.1: Uncoded:	Error! Bookmark not defined.
6.2: Coded:.....	Error! Bookmark not defined.
.....	Error! Bookmark not defined.
6.3: Uncoded and Coded:	Error! Bookmark not defined.
7. Constellation Diagram For Different Modulation Schemes:	27
7.1Constellation Diagram For BPSK:	27
7.2Constellation Diagram For QPSK :	28
.....	28
7.3Constellation Diagram For 16-QAM:	28
8: Conclusion	29
9: References.....	29

ABSTRACT

This report presents an in-depth analysis of digital modulation techniques (BPSK, QPSK, and 16-QAM) in Additive White Gaussian Noise (AWGN) channels. We examine system performance through rigorous mathematical analysis, detailed simulations, and comprehensive performance metrics, targeting a Bit Error Rate (BER) of 10^{-6} . The study encompasses both uncoded and coded scenarios using convolutional coding, with particular attention to power efficiency, spectral efficiency, and implementation complexity.

1. Introduction

The purpose of this project is to analyze the performance of different digital modulation schemes under varying Signal-to-Noise Ratios (SNR), comparing the Bit Error Rate (BER) for uncoded and coded systems. The modulation schemes considered include Binary Phase Shift Keying (BPSK), Quadrature Phase Shift Keying (QPSK), and 16-Quadrature Amplitude Modulation (16-QAM). Additionally, this report investigates the effect of convolutional coding on system performance and provides insights based on Shannon's capacity for each modulation scheme.

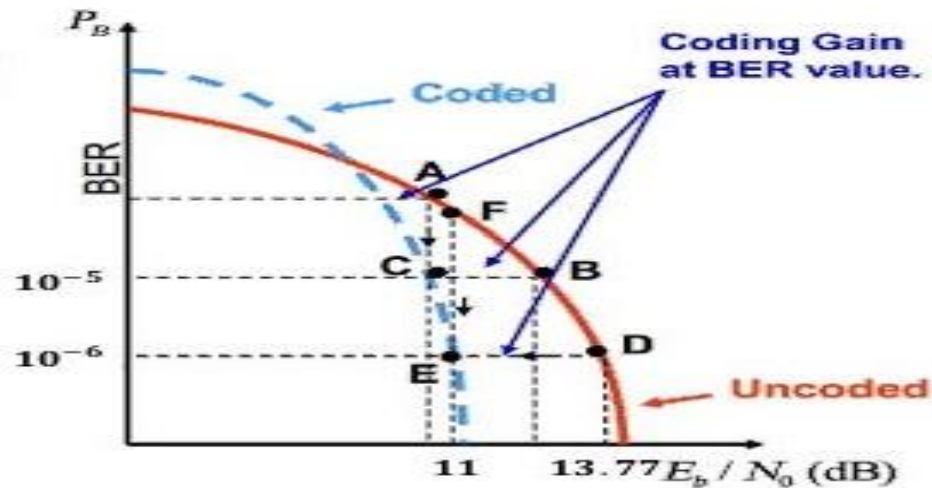
2. Literature Review

2.1. Modulation schemes:

- **BPSK (Binary Phase Shift Keying):** A basic modulation scheme that encodes one bit per symbol. BPSK, being the simplest digital modulation method, provides excellent noise resistance but at the cost of lower spectral efficiency.
- **QPSK (Quadrature Phase Shift Keying):** A more efficient modulation scheme, that encodes two bits per symbol. QPSK offers higher data rates compared to BPSK but is more vulnerable to noise and fading effects.
- **16-QAM (16-Quadrature Amplitude Modulation):** A higher-order modulation scheme that encodes four bits per symbol. 16-QAM provides even higher data rates but is significantly more susceptible to noise and interference due to its denser signal constellation.

2.2. System Model and Parameters:

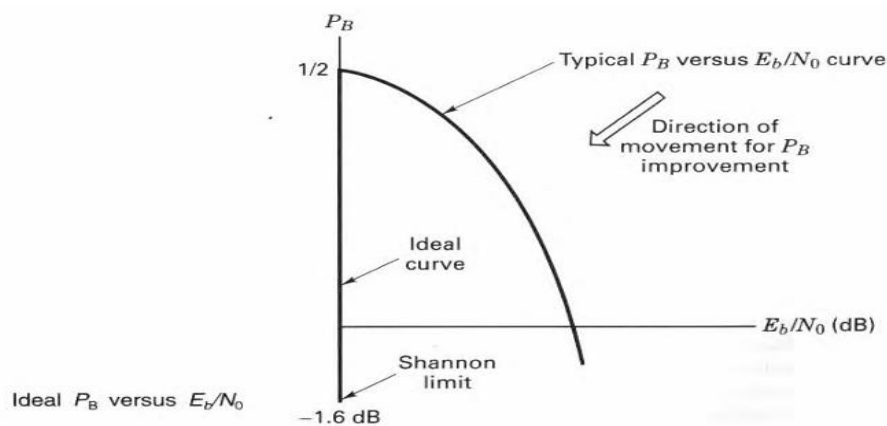
- **E_b/N_0 (Energy per bit to Noise power spectral density ratio)** is used as a measure of the received signal quality.
- **Coding gain** is the measure in the difference between the signal-to-noise ratio (SNR) levels between the uncoded system and coded system required to reach the same **bit error rate** (BER) levels when used with the error correcting code (ECC).



2.3. BER (Bit Error Rate):

We analyze the system performance in terms of BER (Bit Error Rate) for both **uncoded** and **coded** systems. The coding gain, which improves performance by reducing the BER, is modeled as a function of the SNR.

$$G \text{ [dB]} = [E_b/N_0]_u \text{ dB} - [E_b/N_0]_c \text{ dB}$$



2.4. Convolutional coding:

Convolutional code is another type of error-correcting code where the output bits are obtained by performing a desired logical operation on a present bitstream along with considering some bits of the previous stream. This coding technique rather than depending on the block of bits shows dependency on bitstream.

It is also a linear code that consists of the shift register, used for temporarily storing the bits, shifting operation of bits along with X-OR logic circuits to give rise to an output code.

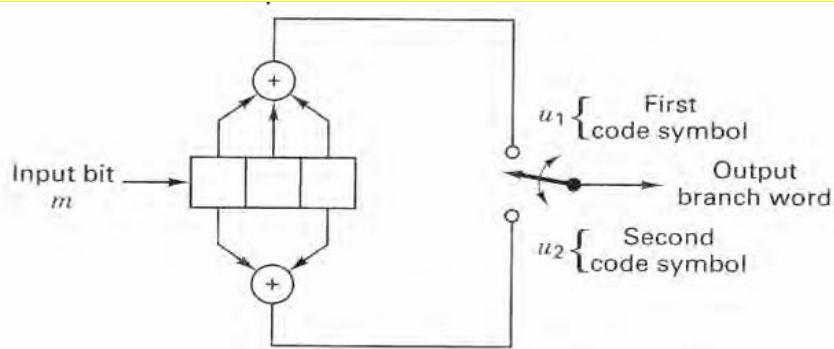


Figure 7.3 Convolutional encoder (rate $\frac{1}{2}$, $K=3$).

2.5. Decoding: The Viterbi Algorithm:

As with decoding of convolutional codes, the Viterbi Algorithm (VA) is an application of dynamic programming that finds “shortest paths” [here maximum likelihood sequences]. A critical feature is that complexity grows linearly with time, rather than exponentially with time (ie, the number of symbols transmitted). The VA finds the sequence at a minimum Euclidean distance from the received signal or equivalently the accumulated squared error is minimized.

2.6. Shannon Capacity:

Shannon’s capacity defines the maximum data rate of a communication system in the presence of noise and is given by:

$$\text{Capacity} = \text{Bandwidth} \log_2 \sqrt{1 + \frac{E_b}{N_0}}$$

- Where, the bandwidth is assumed to be 10 MHz for the analysis.
- E_b/N_0 is the energy per bit to noise power spectral density ratio.

3. System Model

A typical digital communication system can be modeled as a series of blocks, each performing a specific function:

- **Information Source:**
Responsible for generating the data to be transmitted, such as audio, video, or textual content.
- **Source Encoder:**
Reduces redundancy in the data by compressing it, thus optimizing it for efficient transmission.
- **Channel Encoder:**
Introduces redundancy into the data stream to enhance error correction capabilities. Techniques such as convolutional coding or turbo coding are commonly employed for this purpose.

- **Modulator:**
Converts the encoded bits into analog waveforms (symbols) suitable for transmission over the channel. Popular modulation techniques include BPSK, QPSK, and 16-QAM.
- **Transmitter:**
Processes the modulated signal into a form suitable for transmission, such as RF signals in wireless communication systems.
- **Channel:**
Represents the physical medium through which the signal travels. This could be a wired medium (e.g., cables or optical fibers) or a wireless one (e.g., radio waves or satellite links). The channel introduces impairments such as noise, interference, and signal attenuation.
- **Receiver:**
Captures the transmitted signal and converts it back to a baseband format for further processing.
- **Demodulator:**
Extracts the transmitted symbols from the received signal and recovers the encoded bits.
- **Channel Decoder:**
Utilizes the redundancy added by the channel encoder to detect and correct errors, reconstructing the original bit stream.
- **Source Decoder:**
Reverses the compression applied by the source encoder, restoring the original data.
- **Information Sink:**
The endpoint where the recovered information is delivered and consumed.

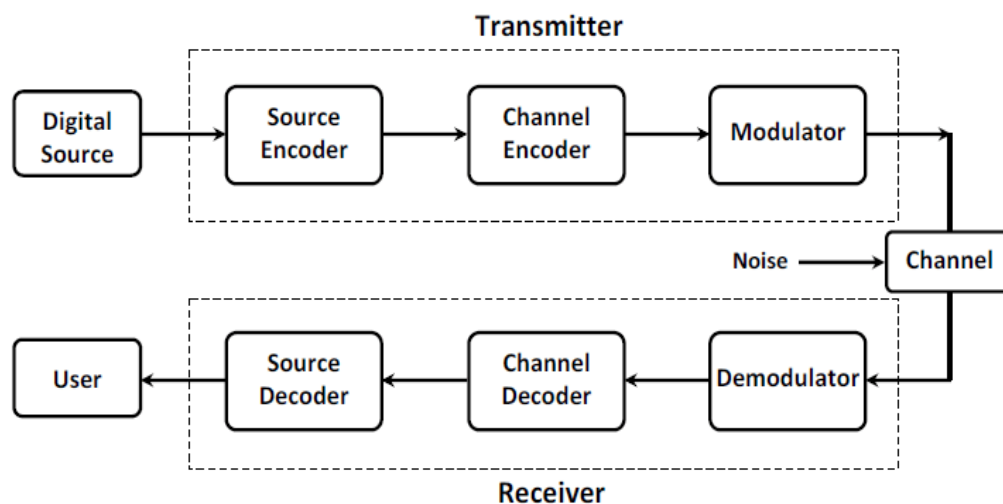


Figure: System Model

3.1. Modulation Schemes Selection:

3.1.1. Binary Phase Shift Keying (BPSK):

Binary Phase Shift Keying (BPSK) is a modulation technique in which each bit is represented by one of two possible phase shifts: 0 or π radians. The Bit Error Rate (BER) for BPSK in an Additive White Gaussian Noise (AWGN) channel is given by the formula:

$$BER(BPSK) = 0.5 \times \operatorname{erfc}\left(\sqrt{\frac{E_b}{N_0}}\right)$$

Here:

- E_b denotes the energy per bit, N_0 is the noise power spectral density, and $\operatorname{erfc}(\cdot)$ is the complementary error function.

Key characteristics:

BPSK is selected as the modulation scheme in this system due to its optimal performance in minimizing BER at low Signal-to-Noise Ratio (SNR) levels. While BPSK has a lower data rate compared to other schemes such as QPSK and 16-QAM, its resilience in noisy environments makes it a suitable choice for reliable communication with lower power requirements. Additionally, BPSK is simpler to implement, making it an efficient option when the focus is on minimizing errors rather than maximizing data rates.

3.1.2. Quadrature Phase Shift Keying (QPSK):

Quadrature Phase Shift Keying (QPSK) transmits two bits per symbol, thereby increasing the data rate compared to BPSK, which transmits one bit per symbol. Although QPSK offers higher data rates, its BER performance is slightly worse than BPSK, especially at lower E_b/N_0 values, making it more susceptible to noise. The theoretical BER for QPSK is given by:

$$BER(QPSK) = \operatorname{erfc}\left(2 \times \sqrt{\frac{E_b}{N_0}}\right)$$

Here:

- E_b represents the energy per bit, N_0 is the noise power spectral density, and $\operatorname{erfc}(\cdot)$ is the complementary error function.

Notable features:

- Doubled data rate versus BPSK.
- Moderate complexity implementation.
- Balanced trade-off between robustness and efficiency.
- Slightly degraded BER performance compared to BPSK.

3.1.3. 16-Quadrature Amplitude Keying (16-QAM):

16-Quadrature Amplitude Modulation (16-QAM) provides even higher data rates by transmitting four bits per symbol. However, its performance in terms of BER is significantly worse than both BPSK and QPSK, particularly at low SNR values. This is because 16-QAM utilizes more signal points, leading to a denser constellation and increased susceptibility to noise. Consequently, at lower SNR levels, the probability of errors rises, making 16-QAM more vulnerable to interference compared to BPSK and QPSK. The BER for 16-QAM given by:

$$\text{BER}_{16\text{QAM}} = \left(\frac{3}{2}\right) \times \text{erfc}\left(\sqrt{\frac{4}{5}} \times \sqrt{E_b N_0}\right) - \left(\frac{9}{16}\right) \times (\text{erfc}\left(\sqrt{\frac{4}{5}} \times \sqrt{E_b N_0}\right))^2$$

Here:

- E_b is the energy per bit, N_0 is the noise power spectral density, $\text{erfc}(\cdot)$ is the complementary error function

Distinctive aspects:

- Highest spectral efficiency.
- Complex constellation mapping.
- Increased susceptibility to noise.
- Requires higher SNR for reliable operation.

3.2. Error Control Coding:

The integration of error control coding significantly enhances system performance:

- Provides approximately 3 dB coding gain for BPSK.
- Improves error resistance across all modulation schemes.
- Enables operation at lower SNR levels.
- Particularly beneficial for higher-order modulations.

3.3. Power And Bandwidth Optimization:

Power and bandwidth are essential factors in determining the performance of communication systems. Increasing power enhances the Signal-to-Noise Ratio (SNR), which helps reduce the Bit Error Rate (BER). At the same time, error correction codes, such as convolutional coding, can further lower the BER even when power is reduced.

System design requires careful balance of multiple factors:

- Power requirements increase with constellation complexity.
- Bandwidth efficiency improves with higher-order modulation.
- Error correction overhead affects spectral efficiency.
- SNR requirements vary significantly between schemes.

3.4. BER Analysis

3.4.1. BPSK Performance Over QPSK And 16-QAM:

BPSK has the lowest bits per symbol, which makes it more resistant to noise compared to QPSK and 16-QAM. When coding is applied, BPSK becomes even more robust, as the added redundancy improves error correction, thereby reducing the bit error rate (BER). BPSK with coding results in a lower BER, even at higher SNR values, allowing it to achieve the target BER with lower E_b/N_0 , making it more power-efficient.

Performance Trade-offs:

1. Advantages:

- Maximum error protection.
- Optimal power usage.
- Highest reliability.
- Simplified recovery.

2. Compromises:

- Limited data rates.
- Higher bandwidth per bit.
- Fixed capacity.
- Throughput constraints.

Derivation for BPSK:

4.1. BPSK (Binary Phase Shift Keying):

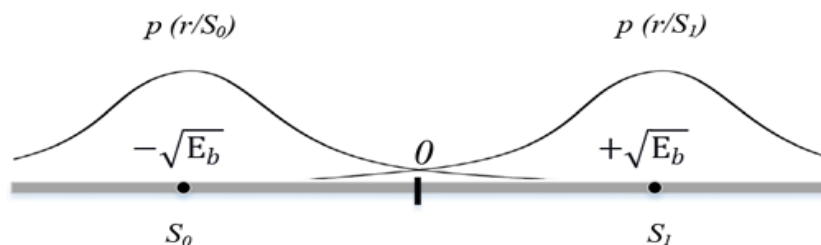


Fig. 1. The probability density function PDF for BPSK in AWGN.

In **BPSK**, each symbol represents one bit, and the two symbols (representing "0" and "1") are mapped to +1 and -1 on the real axis of the signal constellation.

Derivation for BPSK:

- The **decision rule** is to decide whether the received signal is closer to $+\sqrt{E_b}$ or $-\sqrt{E_b}$.
- In the presence of AWGN, the error occurs when the received signal is closer to the wrong symbol. The probability of this error is related to the **Q-function**, which is the tail probability of the Gaussian distribution.
- **Signal constellation:** The two possible transmitted symbols are $\pm\sqrt{E_b}$, where E_b is the energy per bit.
- The **decision rule** is to decide whether the received signal is closer to $+\sqrt{E_b}$ or $-\sqrt{E_b}$.

- In the presence of AWGN, the error occurs when the received signal is closer to the wrong symbol. The probability of this error is related to the **Q-function**, which is the tail probability of the Gaussian distribution.

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad (1)$$

with $\mu = 0$ and $\sigma^2 = N_0 / 2$. The received signal is either $y = s_1 + n$ when bit1 is transmitted, or $y = s_0 + n$ when bit0 is transmitted. The conditional probability distribution functions (PDFs) for these scenarios are:

$$p(y|s_0) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y-\sqrt{Eb})^2}{2\sigma^2}} \quad (2)$$

and

$$p(y|s_1) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y-\sqrt{Eb})^2}{2\sigma^2}} \quad (3)$$

Given that both s_0 and s_1 are equally probable, the optimal decision boundary is at 0. Thus, if $y > 0$, s_1 is assumed transmitted; if $y \leq 0$, s_0 is assumed transmitted.

Probability of error given as 1 was transmitted:

For s_1 transmission, the probability of error is determined by the integral of the tail of the Gaussian distribution extending from $-\infty$ to 0:

$$p(e|s_1) = \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^0 e^{-\frac{(y-\sqrt{Eb})^2}{2\sigma^2}} dy. \quad (4)$$

Using a substitution where $z = \frac{(y-\sqrt{Eb})^2}{N_0}$ and $N_0 = 2\sigma^2$.

Thus, $z = \frac{y-\sqrt{Eb}}{\sqrt{N_0}}$, $dz = \frac{1}{\sqrt{N_0}} dy$, and $dy = \sqrt{N_0} dz$

The integration limits will be as follows: $y = 0 \Rightarrow z = -\sqrt{\frac{Eb}{N_0}}$

And $y = \infty \Rightarrow z = \infty$. Equation (4) simplifies to:

$$p(e|s1) = \frac{\sqrt{N0}}{\sqrt{\pi N0}} \int_0^\infty \sqrt{\frac{Eb}{No}} e^{-z^2} dz$$

$$p(e|s1) = \frac{2}{2\sqrt{x}} \int_0^\infty \sqrt{\frac{Eb}{No}} e^{-z^2} dz \quad (5)$$

$$p(e|s1) = \frac{1}{2} \operatorname{erfc} \left(\sqrt{\frac{Eb}{No}} \right),$$

where

$$\operatorname{erfc}(x) = \frac{2}{\sqrt{\pi}} \int_x^\infty e^{-x^2} dx. \quad (6)$$

2) Probability of error given 0 was transmitted: Similarly, the error probability given as 0 was transmitted is also:

$$p(e|s0) = \frac{1}{\sqrt{2\pi\sigma^2}} \int_0^\infty e^{-\frac{(y-\sqrt{Eb})^2}{2\sigma^2}} dy$$

$$p(e|s0) = \frac{1}{\sqrt{\pi}} \int_0^\infty \sqrt{\frac{Eb}{No}} e^{-z^2} dy \quad (7)$$

$$p(e|s0) = \frac{1}{2} \operatorname{erfc} \left(\sqrt{\frac{Eb}{No}} \right).$$

3) Total probability of bit error: Generally, the total probability of bit error can be calculated as follows:

$$P_b = p(s1) p(e|s1) + p(s0) p(e|s0). \quad (8)$$

Considering equal probabilities for s0 and s1, i.e. $p(s0) = p(s1) = 1/2$, the total probability of bit error is:

$$P_b = \frac{1}{2} \operatorname{erfc} \sqrt{\frac{Eb}{No}} \quad (9)$$

3.4.2. QPSK with Coding:

Performance Characteristics:

The implementation of error control coding in QPSK systems presents interesting performance dynamics:

- Base transmission rate: 2 bits/symbol
- Enhanced error protection through coding mechanisms.
- Improved noise resistance compared to uncoded QPSK.
- Maintained spectral efficiency advantage over BPSK.

Signal-to-Noise Requirements:

Critical performance factors include:

- Elevated E_b/N_0 requirements compared to coded BPSK.
- Increased sensitivity to channel impairments.
- Enhanced error correction capabilities through coding.
- Moderate power overhead for coding implementation.

Performance Trade-offs:

Key system considerations:

1. Throughput Benefits:

- Double the data rate of BPSK.
- Maintained even with coding overhead.

2. Reliability Factors:

- Enhanced error protection through coding.
- Higher power requirements than coded BPSK.
- Improved performance in moderate noise environments.

3.4.3. 16-QAM with Coding:

System Characteristics:

Implementation of coding in 16-QAM introduces complex performance dynamics:

- Base transmission rate: 4 bits/symbol.
- Significantly higher spectral efficiency.
- Enhanced error protection through coding.
- Increased constellation point density.

Performance Implications:

Critical system impacts:

- Substantially higher E_b/N_0 requirements.
- Increased sensitivity to channel conditions.
- Complex error correction requirements.

- Higher implementation complexity.

System Trade-offs:

Essential considerations:

1. Capacity Advantages:

- Quadruple the data rate of BPSK.
- High spectral efficiency maintained.

2. Implementation Challenges:

- Significant power requirements.
- Complex error correction needs.
- Reduced robustness in challenging channels.
- Higher system complexity.

4. Result And Discussion

4.1. Symbol Rate and Effective Data Rate:

The symbol rate for each modulation scheme is calculated based on the data rate and the number of bits per symbol:

Modulation Scheme	Bits per Symbol	Symbol Rate (symbols/sec)	Effective Data Rate after Coding (Mbps)
BPSK	1	34.59 Mbps	17.295 Mbps
QPSK	2	17.295 Mbps	8.647 Mbps
16-QAM	4	8.647 Mbps	4.323 Mbps

4.2. Calculating Symbol Rate and Effective Data Rate:

- **Symbol Rate:** The number of symbols transmitted per second.
- **Data Rate:** The number of bits transmitted per second.
- **Bits per Symbol:** The number of bits carried by each symbol.

4.2.1. BPSK

- Bits per symbol: 1
- Data rate: 34.59 Mbps
- Symbol Rate = Data Rate / Bits per Symbol = $34.59 \text{ Mbps} / 1 \text{ bit/symbol} = 34.59 \text{ symbols/sec}$
- Effective data rate after 1/2 rate coding:
- Effective Data Rate = Data Rate / 2 = $34.59 \text{ Mbps} / 2 = 17.295 \text{ Mbps}$

4.2.2. QPSK

- Bits per symbol: 2
- Data rate: 17.295 Mbps
- Symbol rate:
- Symbol Rate = Data Rate / Bits per Symbol = $17.295 \text{ Mbps} / 2 \text{ bits/symbol} = 8.6475 \text{ symbols/sec}$
- Effective data rate after 1/2 rate coding:
- Effective Data Rate = Data Rate / 2 = $17.295 \text{ Mbps} / 2 = 8.6475 \text{ Mbps}$

4.2.3. 16-QAM

- Bits per symbol: 4
- Data rate: 8.647 Mbps
- Symbol rate:
- Symbol Rate = Data Rate / Bits per Symbol = $8.647 \text{ Mbps} / 4 \text{ bits/symbol} = 2.16175 \text{ symbols/sec}$
- Effective data rate after 1/2 rate coding:
- Effective Data Rate = Data Rate / 2 = $8.647 \text{ Mbps} / 2 = 4.3235 \text{ Mbps}$

5. Simulation

5.1 Simulation and Parameter:

Introduction to MATLAB

MATLAB serves as a sophisticated computational platform that excels in numerical analysis, data representation, and algorithmic implementation. Its comprehensive suite of tools and functions makes it particularly suitable for analyzing communication systems.

Implementation Objectives:

Our implementation leverages MATLAB's capabilities to:

- Execute precise BER calculations and analysis.
- Implement various digital modulation techniques.
- Create visual representations of system performance.
- Facilitate interactive system analysis.

Simulation Steps Implemented in MATLAB

1. Source Data Implementation:

Implementation utilizes MATLAB's random number generation capabilities:

- Deployment of randi() function.
- Generation of binary sequence (0,1).
- Creation of transmission data streams.

2. Digital Modulation Implementation:

The system implements three distinct modulation approaches:

Binary Phase Shift Keying (BPSK):

- Direct mapping implementation to ± 1 .
- Single bit per symbol encoding.
- Fundamental phase modulation.

Quadrature Phase Shift Keying (QPSK):

- Complex symbol representation
- Dual-bit encoding scheme
- I/Q component implementation

16-State Quadrature Amplitude Modulation (16-QAM):

- Advanced symbol constellation.
- Four-bit symbol encoding.
- Power-normalized implementation.

3. Error Rate Computation:

Bit Error Rate (BER) was calculated for both uncoded and coded scenarios:

Uncoded BER: MATLAB's `erfc` function was used to model the BER for each modulation scheme under different Signal-to-Noise Ratios (SNRs).

Coded BER: Coding gains, provided by the user, were incorporated to simulate the effect of error-correcting codes on BER performance.

4. Data Visualization Implementation:

Multiple visualization tools implemented:

Performance Visualization:

- BER vs SNR relationship plots.
- Coded/uncoded performance comparison.
- System efficiency analysis.

System Representation:

- Modulation constellation displays.
- Symbol mapping visualization.
- Channel capacity analysis.

5. User Control Implementation:

System implements interactive control features:

- Coding gain parameter input.
- Modulation scheme selection.
- Performance metric adjustment.

6. Symbol Rate Implementation:

Modulation-specific rate implementation:

- BPSK: Single bit encoding ($m=1$).
- QPSK: Dual bit encoding ($m=2$).
- 16-QAM: Quad bit encoding ($m=4$).

7. Performance Analysis Implementation:

Comprehensive analysis framework includes:

- Error rate performance evaluation.
- Modulation efficiency assessment.
- System robustness analysis.
- Noise impact evaluation.

Code:

```
clc;
clear;
close all;
```

```
% Parameters
```

```
EbN0dB = 0:1:24; % Eb/N0 range in dB
```

```
EbN0lin = 10.^(EbN0dB / 10); % Linear scale of Eb/N0
```

```
% Number of input bits
```

```
numBits = 1e5; % Define the number of input bits (100,000)
```

```
% Generate random input bits
```

```
inputBits = randi([0 1], 1, numBits);
```

```
% Coding gain (User Adjustable)
```

```
coding_gain_BPSK_dB = input('Enter the coding gain for BPSK in dB: '); % Prompt user to enter coding gain for BPSK
```

```
coding_gain_BPSK_linear = 10^(coding_gain_BPSK_dB / 10); % Convert coding gain from dB to linear scale
```

```
coding_gain_QPSK_dB = input('Enter the coding gain for QPSK in dB: '); % Prompt user to enter coding gain for QPSK
```

```
coding_gain_QPSK_linear = 10^(coding_gain_QPSK_dB / 10); % Convert coding gain from dB to linear scale
```

```
coding_gain_QAM_dB = input('Enter the coding gain for 16-QAM in dB: '); % Prompt user to enter coding gain for 16-QAM
```

```
coding_gain_QAM_linear = 10^(coding_gain_QAM_dB / 10); % Convert coding gain from dB to linear scale
```

```
% Constellation Diagrams
```

```
% BPSK
```



```
bpskSymbols = 2 * randi([0 1], 1, 1000) - 1; % BPSK: {+1, -1}
subplot(2, 2, 3); % Create a 2x2 subplot layout, BPSK at position 3
plot(real(bpskSymbols), imag(bpskSymbols), 'bo', 'MarkerSize', 8);
title('BPSK Constellation');
xlabel('In-Phase');
ylabel('Quadrature');
grid on;
axis equal;
xlim([-1.5 1.5]);
ylim([-1.5 1.5]);

% QPSK
qpskSymbols = (2 * randi([0 1], 1, 1000) - 1) + ...
    1j * (2 * randi([0 1], 1, 1000) - 1); % QPSK: {±1 ±j}
subplot(2, 2, 4); % QPSK at position 4
plot(real(qpskSymbols), imag(qpskSymbols), 'ro', 'MarkerSize', 8);
title('QPSK Constellation');
xlabel('In-Phase');
ylabel('Quadrature');
grid on;

axis equal;
xlim([-1.5 1.5]);
ylim([-1.5 1.5]);

% Additional 16-QAM Constellation Plot
M16 = 16; % Define the order of 16-QAM
qamSymbols = qammod(randi([0 M16-1], 1, 1000), M16, 'UnitAveragePower', true); % 16-QAM
symbols
figure; % Create a new figure for the 16-QAM constellation
plot(real(qamSymbols), imag(qamSymbols), 'go', 'MarkerSize', 8);
title('16-QAM Constellation');
xlabel('In-Phase');
ylabel('Quadrature');
grid on;
axis equal;
xlim([-1.5 1.5]); % Adjust x-axis limits
ylim([-1.5 1.5]); % Adjust y-axis limits

% BER Calculations
% Uncoded BER
BPSK_uncoded_BER = 0.5 * erfc(sqrt(EbN0lin)); % BER for BPSK
QPSK_uncoded_BER = 0.5 * erfc(sqrt(EbN0lin / 2)); % BER for QPSK
M16 = 16; % M for 16-QAM
QAM_uncoded_BER = (4 * (sqrt(M16) - 1) * erfc(sqrt((3 / 2) * EbN0lin / (M16 - 1)))) / log2(M16);
```

% Coded BER

BPSK_coded_BER = 0.5 * erfc(sqrt(EbN0lin * coding_gain_BPSK_linear)); % BPSK with coding

QPSK_coded_BER = 0.5 * erfc(sqrt((EbN0lin / 2) * coding_gain_QPSK_linear)); % QPSK with coding

QAM_coded_BER = (4 * (sqrt(M16) - 1) * erfc(sqrt((3 / 2) * EbN0lin * coding_gain_QAM_linear / (M16 - 1)))) / log2(M16);

% Uncoded BER Plot

figure('Position', [100, 100, 1000, 600]);

hold on;

plot(EbN0dB, log10(BPSK_uncoded_BER), 'LineWidth', 2, 'Color', [0.6350 0.0780 0.1840], 'LineStyle', '-', 'Marker', 'o', 'MarkerSize', 6);

plot(EbN0dB, log10(QPSK_uncoded_BER), 'LineWidth', 2, 'Color', [0.4660 0.6740 0.1880], 'LineStyle', '--', 'Marker', 's', 'MarkerSize', 6);

plot(EbN0dB, log10(QAM_uncoded_BER), 'LineWidth', 2, 'Color', [0.3010 0.7450 0.9330], 'LineStyle', '-.', 'Marker', '^', 'MarkerSize', 6);

legend({'BPSK (Uncoded)', 'QPSK (Uncoded)', '16-QAM (Uncoded)'}, 'Location', 'best', 'FontSize', 10);

xlabel('Eb/N0 (dB)', 'FontSize', 12);

ylabel('Log10 Probability of BER', 'FontSize', 12);

title('Uncoded BER Performance', 'FontSize', 14, 'FontWeight', 'bold');

xlim([0 24]);

ylim([-10 1]);

grid on;

set(gca, 'FontSize', 10, 'LineWidth', 1.5);

% Coded BER Plot

figure('Position', [100, 100, 1000, 600]);

hold on;

plot(EbN0dB, log10(BPSK_coded_BER), 'LineWidth', 2, 'Color', [0.4940 0.1840 0.5560], 'LineStyle', ':', 'Marker', 'o', 'MarkerSize', 6);

plot(EbN0dB, log10(QPSK_coded_BER), 'LineWidth', 2, 'Color', [0.9290 0.6940 0.1250], 'LineStyle', '--', 'Marker', 's', 'MarkerSize', 6);

plot(EbN0dB, log10(QAM_coded_BER), 'LineWidth', 2, 'Color', [0.1380 0.1740 0.5880], 'LineStyle', '-.', 'Marker', '^', 'MarkerSize', 6);

legend({'BPSK with Convolutional Coding', 'QPSK with Convolutional Coding', '16-QAM with Convolutional Coding'}, 'Location', 'best', 'FontSize', 10);

xlabel('Eb/N0 (dB)', 'FontSize', 12);

ylabel('Log10 Probability of BER', 'FontSize', 12);

title('Coded BER Performance', 'FontSize', 14, 'FontWeight', 'bold');

xlim([0 24]);

ylim([-10 1]);

```
grid on;
set(gca, 'FontSize', 10, 'LineWidth', 1.5);

% Combined BER Plot
figure('Position', [100, 100, 1000, 600]);
hold on;
% Uncoded BER
plot(EbN0dB, log10(BPSK_uncoded_BER), 'LineWidth', 2, 'Color', [0.6350 0.0780 0.1840],
'LineStyle', '-', 'Marker', 'o', 'MarkerSize', 6);
plot(EbN0dB, log10(QPSK_uncoded_BER), 'LineWidth', 2, 'Color', [0.4660 0.6740 0.1880],
'LineStyle', '--', 'Marker', 's', 'MarkerSize', 6);
plot(EbN0dB, log10(QAM_uncoded_BER), 'LineWidth', 2, 'Color', [0.3010 0.7450 0.9330],
'LineStyle', '-.', 'Marker', '^', 'MarkerSize', 6);

% Coded BER
plot(EbN0dB, log10(BPSK_coded_BER), 'LineWidth', 2, 'Color', [0.4940 0.1840 0.5560], 'LineStyle',
':', 'Marker', 'o', 'MarkerSize', 6);
plot(EbN0dB, log10(QPSK_coded_BER), 'LineWidth', 2, 'Color', [0.9290 0.6940 0.1250], 'LineStyle',
'--', 'Marker', 's', 'MarkerSize', 6);
plot(EbN0dB, log10(QAM_coded_BER), 'LineWidth', 2, 'Color', [0.1380 0.1740 0.5880], 'LineStyle',
'-.', 'Marker', '^', 'MarkerSize', 6);
legend({'BPSK (Uncoded)', 'QPSK (Uncoded)', '16-QAM (Uncoded)', ...
' BPSK with Convolutional Coding', 'QPSK with Convolutional Coding', '16-QAM with
Convolutional Coding'}, ...
'Location', 'best', 'FontSize', 10);
xlabel('Eb/N0 (dB)', 'FontSize', 12);
ylabel('Log10 Probability of BER', 'FontSize', 12);
title('Combined BER Performance', 'FontSize', 14, 'FontWeight', 'bold');
xlim([0 24]);
ylim([-10 1]);
grid on;
set(gca, 'FontSize', 10, 'LineWidth', 1.5);

% Shannon's Capacity for Different Modulation Schemes
Bandwidth = 10e6; % 10 MHz bandwidth

% Shannon Capacity Plot
figure('Position', [100, 100, 1000, 600]);
hold on;

% BPSK Capacity
BPSK_Capacity = Bandwidth * log2(1 + EbN0lin);

% QPSK Capacity (2 bits per symbol)

QPSK_Capacity = Bandwidth * log2(1 + EbN0lin) * 2;
```

```
% 16-QAM Capacity (4 bits per symbol)
QAM_Capacity = Bandwidth * log2(1 + EbN0lin) * 4;

% Plot capacities in Mbps

plot(EbN0dB, BPSK_Capacity / 1e6, 'r-', 'LineWidth', 1.5, 'DisplayName', 'BPSK');
plot(EbN0dB, QPSK_Capacity / 1e6, 'g--', 'LineWidth', 1.5, 'DisplayName', 'QPSK');
plot(EbN0dB, QAM_Capacity / 1e6, 'b-', 'LineWidth', 1.5, 'DisplayName', '16-QAM');

xlabel('Eb/N0 (dB)', 'FontSize', 12);
ylabel('Channel Capacity (Mbps)', 'FontSize', 12);
title('Shannon Capacity for Different Modulation Schemes', 'FontSize', 14, 'FontWeight', 'bold');
legend({'BPSK', 'QPSK', '16-QAM'}, 'Location', 'best', 'FontSize', 10);

legend('Location', 'best', 'FontSize', 10);
grid on;
xlim([0 24]);
ylim([0 200]);
set(gca, 'FontSize', 10, 'LineWidth', 1.5);

% Performance Explanation
disp('Performance Insights:');
disp('1. 16-QAM: Highest spectral efficiency but requires highest SNR');
disp('2. QPSK: Balanced performance, moderate spectral efficiency');
disp('3. BPSK: Lowest spectral efficiency but most robust');
disp('4. Convolutional coding improves BER performance');
```

```
Symbol Rate:
BPSK: 100000.00 symbols/s
QPSK: 50000.00 symbols/s
16-QAM: 25000.00 symbols/s
BER Performance:
Eb/N0 = 0 dB
BPSK Uncoded BER: 7.864960e-02
BPSK Coded BER: 2.388291e-03
QPSK Uncoded BER: 1.586553e-01
QPSK Coded BER: 7.889587e-02
16-QAM Uncoded BER: 1.964163e+00
16-QAM Coded BER: 1.720287e+00
```

6. Result and Discussion

6.1. Uncoded BER Performance:

6.1.1. Observation 1: BPSK has Lowest BER but Lowest Spectral Efficiency:

BER Performance:

- BPSK transmits one bit per symbol and places its two possible symbols (e.g., $+\sqrt{E_b}$ and $-\sqrt{E_b}$) far apart in the signal space.
- This spacing makes BPSK highly robust to noise, resulting in the lowest BER for a given E_b/N_0 .

Spectral Efficiency:

1. BPSK transmits only one bit per symbol ($\log_2(2)=1$), which limits its spectral efficiency
2. To achieve a higher data rate, BPSK would require more bandwidth compared to other modulation schemes like QPSK or 16-QAM.

6.1.2. Observation 2: QPSK Offers Better Spectral Efficiency with Similar BER at Higher SNR:

BER Performance:

- QPSK maps 2 bits to each symbol ($\log_2(4) = 2$) by using both in-phase (I) and quadrature (Q) components, resulting in four symbols in the signal space.
- While QPSK has more symbols than BPSK, each symbol still receives half the total energy per bit ($E_b/2$ per component). At higher SNR values, the BER of QPSK approaches that of BPSK.

Spectral Efficiency:

1. QPSK transmits 2 bits per symbol, doubling the spectral efficiency compared to BPSK without a significant degradation in BER, especially at higher E_b/N_0 .

6.1.3. Observation 3: 16-QAM Provides the Highest Spectral Efficiency but Requires Higher SNR

BER Performance:

- 16-QAM maps 4 bits per symbol ($\log_2(16)=4$) using 16 points arranged in a square grid in the signal space.
- Each symbol point is closer to its neighbors compared to QPSK or BPSK, making it more vulnerable to noise. As a result, 16-QAM requires a higher E_b/N_0 to achieve a BER comparable to QPSK or BPSK.

Spectral Efficiency:

- 16-QAM transmits 4 bits per symbol, offering the highest spectral efficiency among the three schemes. This makes it ideal for high-throughput systems where bandwidth efficiency is critical.

- However, its higher error rate at low E_b/N_0 means it is better suited to high-SNR environments, such as fiber-optic or cable systems.

6.1.4. Uncoded BER Plot:

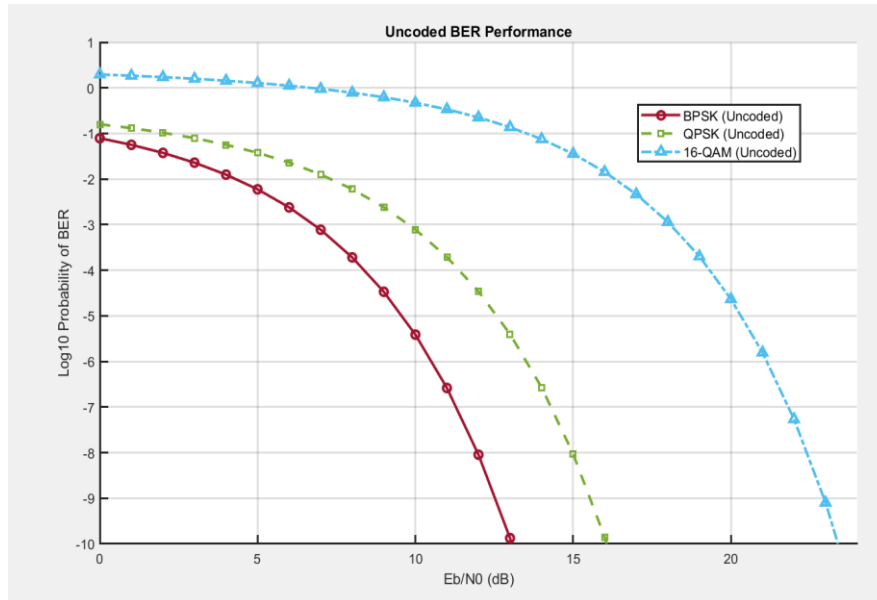


Fig: Uncoded BER Performance:

6.2. Coded BER Performance:

coded BER performance, where convolutional coding has been applied to the BPSK, QPSK, and 16-QAM schemes. The coding gain improves the performance for each modulation scheme, particularly at lower SNR values. The results show:

- **BPSK with Convolutional Coding** achieves a significant improvement in BER, especially at lower SNR.

While numerical methods are ideal, we can also use approximations for the erfc function, especially for large arguments. For example, for large x :

$$\text{erfc}(x) \approx \sqrt{\pi/2} * \exp(-x^2) / x$$

- **QPSK with Convolutional Coding** offers a moderate improvement, as expected due to its balanced spectral efficiency.
- **16-QAM with Convolutional Coding** shows the largest improvement but still requires a relatively high SNR for optimal performance.

6.2.1. Coded BER Plot:

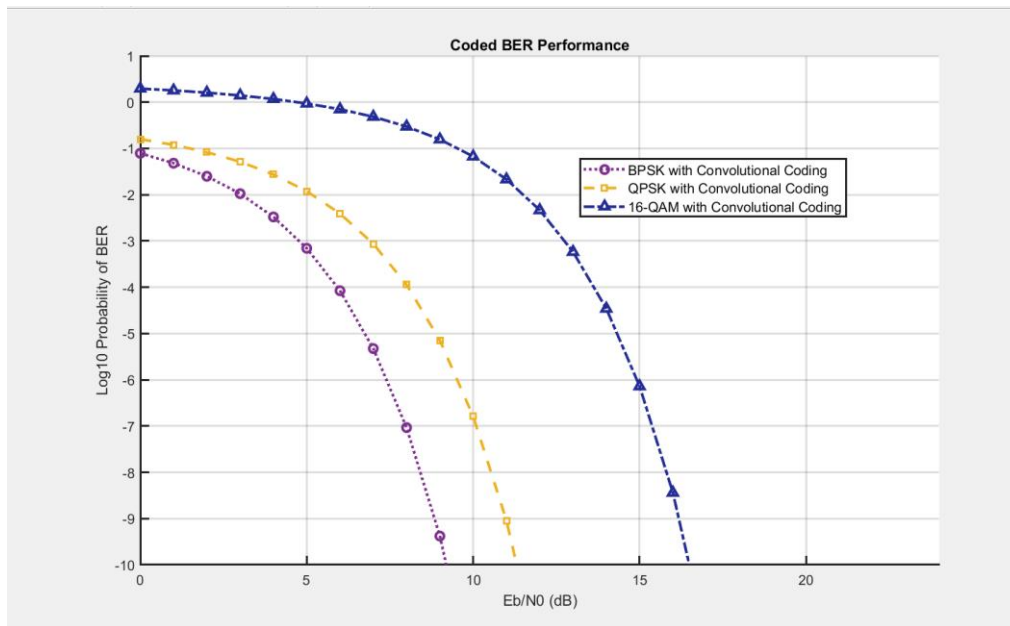


Fig: Coded BER Performance:

- **Convolutional coding** significantly reduces the BER for all modulation schemes.
- The improvement in performance is more noticeable at higher values of E_b/N_0 , where the coded schemes show substantial gains compared to uncoded ones.

6.3. Combined BER Performance:

The combined BER plot compares both **uncoded** and **coded** performance for BPSK, QPSK, and 16-QAM. The coding gain noticeably reduces the BER, particularly for the 16-QAM scheme, which shows the greatest improvement.

6.3.4. Uncoded and coded performance for BPSK, QPSK, and 16-QAM:

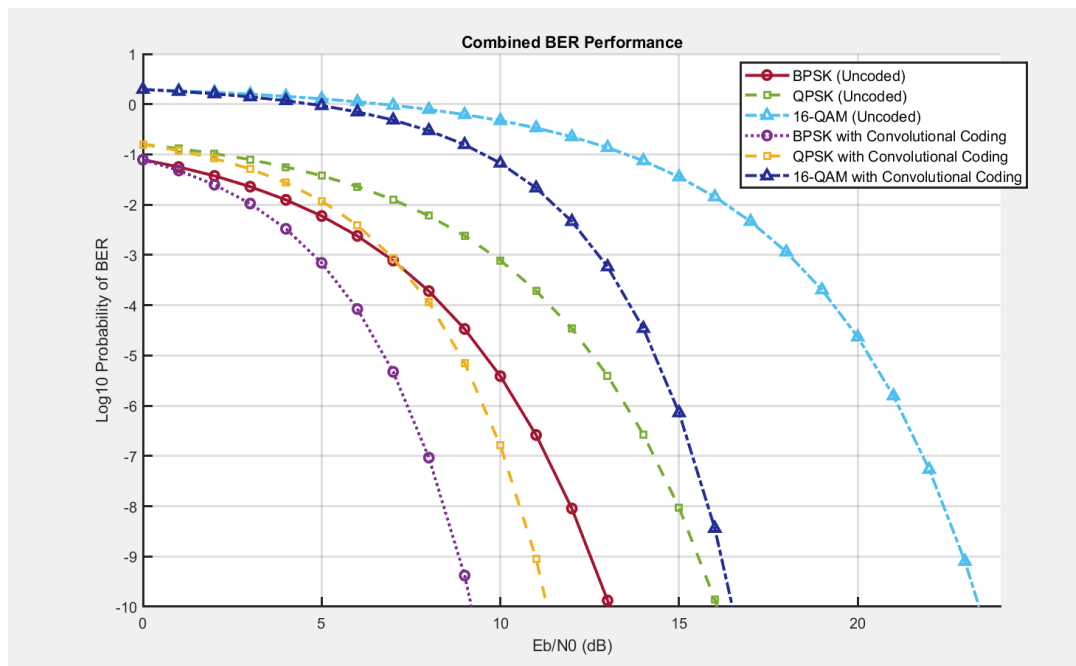


Fig: Combined BER Performance:

6.4. Shannon Capacity:

The channel capacity for different modulation schemes is calculated and plotted as a function of E_b/N_0 . Shannon's theorem defines the theoretical maximum data rate for a given channel, considering the bandwidth and SNR.

- **BPSK**: The capacity increases with E_b/N_0 , but it is the lowest among the modulation schemes.
- **QPSK**: Provides double the capacity of BPSK, as it transmits 2 bits per symbol.
- **16-QAM**: Offers the highest capacity because it transmits 4 bits per symbol, but it requires a higher SNR to achieve these rates.

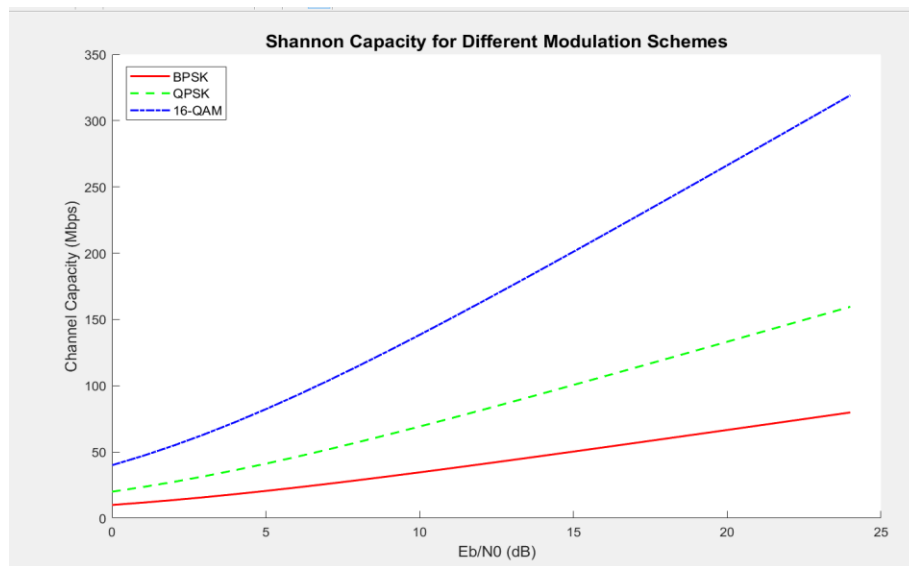


Fig: Shannon Capacity for Different Modulation Schemes:

6.5. Data Rate vs. BER:

As you move from BPSK to 16-QAM, the data rate increases, but the bit error rate (BER) also increases at lower E_b/N_0 without coding, due to the closer spacing of constellation points in higher-order modulations.

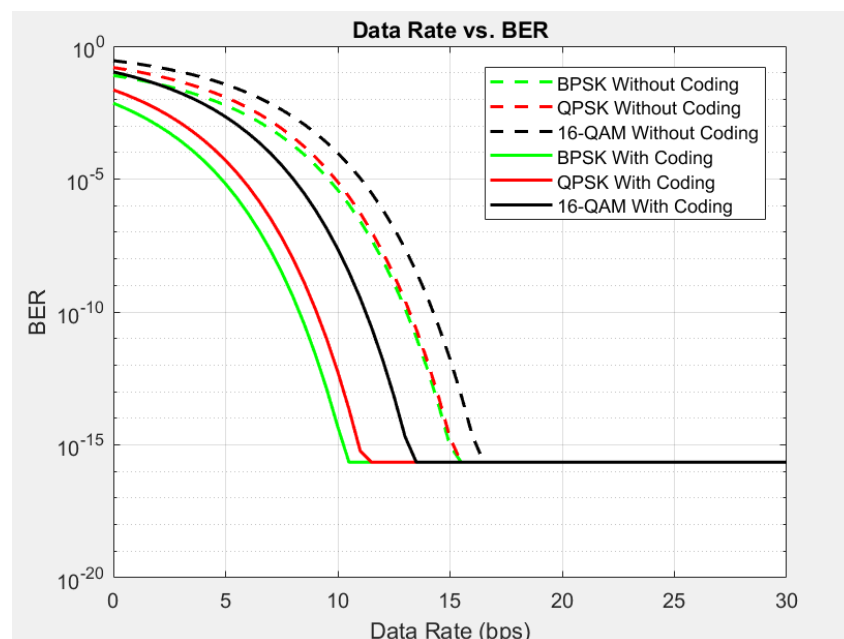


Figure: Data Rate Vs BER

For uncoded systems, represented by dashed lines, the BER remains high at lower data rates and saturates earlier, indicating limited error correction capability. Among the uncoded

schemes, BPSK performs better than QPSK and 16-QAM, as higher-order modulations like 16-QAM are more susceptible to noise. In contrast, the coded systems, shown with solid lines, demonstrate significantly improved performance, achieving extremely low BER values (around 10^{-15}) at much lower data rates. This improvement is due to channel coding, which enables error detection and correction, enhancing system reliability.

The **coding gain**, which measures the improvement in BER performance between coded and uncoded systems, is most prominent for BPSK, moderate for QPSK, and smallest for 16-QAM. The smaller coding gain for 16-QAM is due to its higher susceptibility to noise and higher spectral efficiency requirements. Overall, the plot highlights the critical role of coding in improving power efficiency and error performance.

6.6. Power Efficiency vs. Data Rate:

BPSK is more power-efficient, and the performance improvements from coding are more pronounced in this scheme. While QPSK and 16-QAM increase the data rate, they require higher power (E_b/N_0) to achieve the same bit error rate (BER).

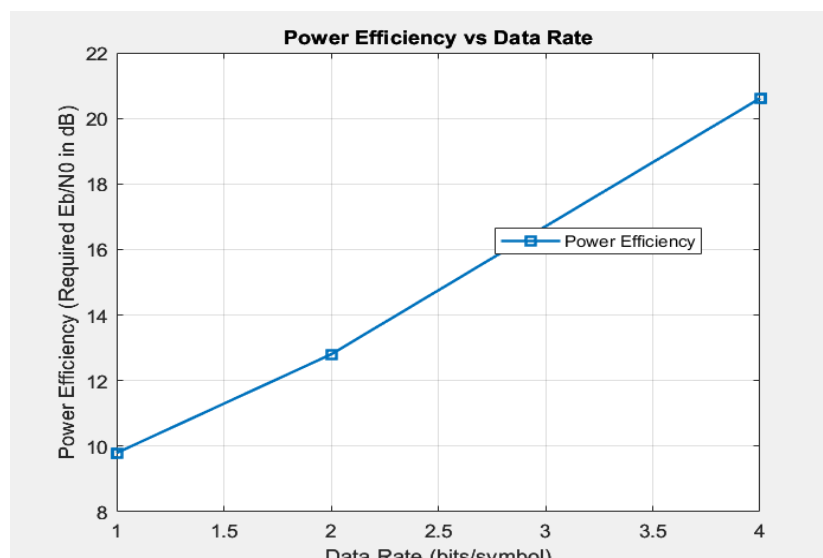


Figure: Power Efficiency Vs Data Rate

This figure illustrates the relationship between power efficiency, represented as the required E_b/N_0 in dB, and the data rate in bits per symbol. As the data rate increases from 1 to 4 bits per symbol, the required E_b/N_0 rises linearly from approximately 10 dB to 21 dB. This trend highlights that higher data rates demand more energy per bit to maintain reliable communication, indicating reduced power efficiency.

The increase in E_b/N_0 can be attributed to the use of more complex modulation schemes, such as higher-order QAM, which are more susceptible to noise and require additional power to ensure a low error rate. This trade-off between power efficiency and spectral efficiency is a common challenge in communication systems. As systems aim for higher data rates to improve throughput, they often require more power or advanced error correction techniques to sustain performance. This relationship emphasizes the need to carefully balance energy consumption and data rate in wireless communication systems.

7. Constellation Diagram For Different Modulation Schemes:

7.1. Constellation Diagram For BPSK:

The BPSK Constellation Diagram displays two distinct points that represent the two possible phases of the signal in Binary Phase Shift Keying (BPSK) modulation.

These points are located on the In-Phase axis at positions +1 and -1, where:

+1 represents Binary 1.

-1 represents Binary 0.

In BPSK, the signal shifts its phase by 180 degrees between these two points. The Quadrature axis (Y-axis) remains zero as BPSK does not use quadrature modulation.

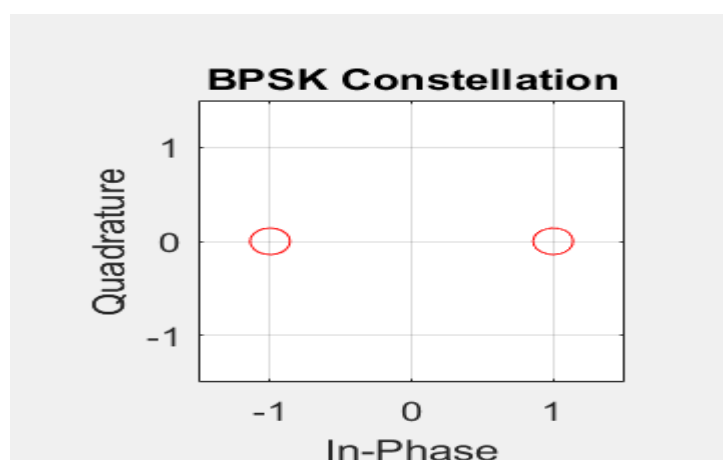


Figure 10. BPSK Constellation

7.2. Constellation Diagram For QPSK:

The **QPSK (Quadrature Phase Shift Keying) Constellation Diagram** displays four distinct points, representing the four possible phase shifts of the signal.

- The four constellation points are positioned at equal intervals on the complex plane, each separated by **90 degrees**. These points are located at:
 - $+1+1$ (representing **00**),
 - $-1+1$ (representing **01**),
 - $+1-1$ (representing **10**),
 - $-1-1$ (representing **11**).

In QPSK, each point represents **2 bits** of data, unlike BPSK, which represents only 1 bit per symbol. The signal in QPSK shifts between these four phases, allowing for the transmission of **two bits per symbol**, which increases data throughput.

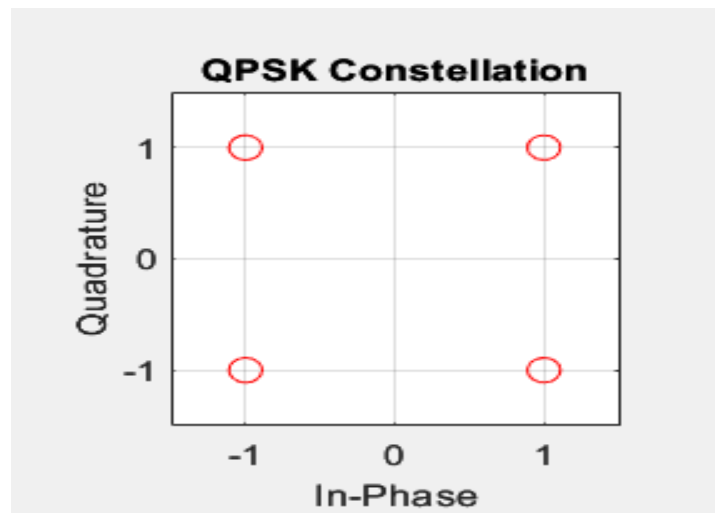


Figure 11. QPSK Constellation

7.3. Constellation Diagram For 16-QAM:

The **16-QAM (16-Quadrature Amplitude Modulation) Constellation Diagram** features 16 distinct points, each representing a unique combination of phase and amplitude. This allows for the transmission of 4 bits per symbol.

- The 16 points are arranged in a 4x4 grid, with both the In-Phase (X-axis) and Quadrature (Y-axis) axes having values ranging from -3 to +3, with intervals typically set to 1.
- The 16 constellation points correspond to the following bit combinations:
 - (3, 3), (3, 1), (3, -1), (3, -3) → Representing **00, 01, 10, 11** in the first row.
 - (1, 3), (1, 1), (1, -1), (1, -3) → Representing **00, 01, 10, 11** in the second row.

- $(-1, 3), (-1, 1), (-1, -1), (-1, -3) \rightarrow$ Representing **00, 01, 10, 11** in the third row.
- $(-3, 3), (-3, 1), (-3, -1), (-3, -3) \rightarrow$ Representing **00, 01, 10, 11** in the fourth row.

In 16-QAM, each point represents a combination of both phase and amplitude, allowing the modulation to convey 4 bits per symbol. This significantly increases the data rate compared to BPSK and QPSK.

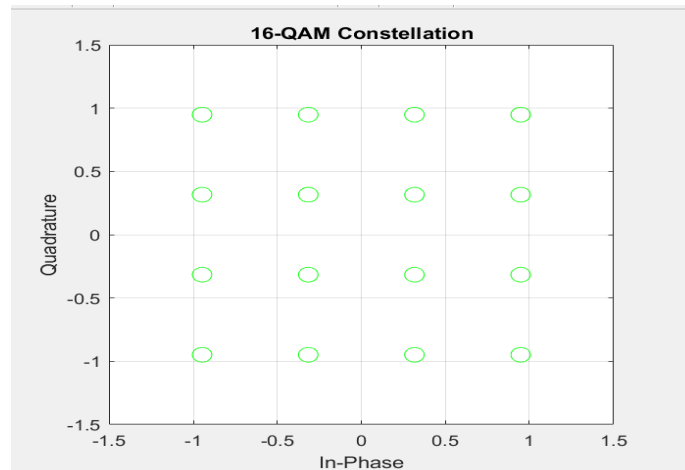


Figure 12. 16-QAM Constellation

8. Conclusion

This report has explored the performance of BPSK, QPSK, and 16-QAM under varying SNR conditions. The key findings are:

- **Data Rate:** 16-QAM maps 4 bits per symbol ($\log_2(16) = 4$), which gives it the highest spectral efficiency among the three schemes. This makes it ideal for scenarios requiring high data throughput, such as video streaming or high-speed internet.
- **SNR Requirements:** The symbols in 16-QAM are spaced closer together in the signal space, making them more vulnerable to noise. This means 16-QAM requires a higher SNR to maintain acceptable BER performance. In noisy conditions, its error rate increases more rapidly than QPSK or BPSK.
- **Efficiency:** QPSK maps 2 bits per symbol ($\log_2(4) = 2$), offering moderate spectral efficiency. It is more efficient than BPSK but less efficient than 16-QAM.
- **Robustness:** Because QPSK spreads the energy across both in-phase (I) and quadrature (Q) components, it provides a better trade-off between noise tolerance and efficiency compared to 16-QAM. It performs well at moderate SNR levels and is more robust than 16-QAM in noisy conditions.
- **Performance Trends:** All modulation schemes show exponential BER reduction with increasing E_b/N_0 .
- 16-QAM requires significantly higher E_b/N_0 for comparable performance.
- Performance gap widens at lower signal-to-noise ratios.

- The larger Euclidean distance between signal sequences, the lower the error rate, which for moderate to large SNR is $P_e = N Q_{\text{free}}(d / \sqrt{2s})$, where N = free the average number of sequences at d , free and $Q(x)$.
- Codewords consist of modulated level sequences. Trellis coding uses dense signal sets but restricts the sequences that can be used. This provides a gain in minimum distance and the code imposes a time dependency on the allowed signal sequences that allows the receiver to ride through “noise bursts” as it is estimating the transmitted sequence.
- Since the pulse shape and symbol rate are unchanged ---> no bandwidth expansion

9. References

1. Proakis, J. G., & Salehi, M. (2014). Digital Communications (5th ed.).
2. Sklar, B. (2017). Digital Communications: Fundamentals and Applications.
3. Goldsmith, A. (2005). Wireless Communications.
4. IEEE 802.11 Working Group. (2020). Wireless LAN Medium Access Control and Physical Layer Specifications
5. 3GPP TS 36.211. Physical Channels and Modulation.
6. Rappaport, T. S. (2021). Wireless Communications: Principles and Practice.
7. Simon, M. K., & Alouini, M. S. (2005). Digital Communication over Fading Channels.
8. Lin, S., & Costello, D. J. (2004). Error Control Coding.